

Algebra (9 th std)

Choose the Appropriate Answer.

1. If $x^3 + 6x^2 + kx + 6$ is exactly divisible by $x + 2$, then $k = 2$

- (a) 6 (b) -7 (c) -8 (d) 11

Ans: (d) 11

$$p(x) = x^3 + 6x^2 + kx + 6 \Rightarrow p(-2) = (-2)^3 + 6(-2)^2 + k(-2) + 6 = 0 \\ \Rightarrow -8 + 24 - 2k + 6 = 0 \Rightarrow 22 - 2k = 0 \Rightarrow k = 22 / 2 = 11$$

2. The root of the polynomial equation $2x + 3 = 0$ is.....

- (a) $1/3$ (b) $-1/3$ (c) $-3/2$ (d) $-2/3$

Ans: (c) $-3/2$

$$2x + 3 = 0 \Rightarrow 2x = -3 \Rightarrow x = -3/2$$

3. The type of the polynomial $4 - 3x^3$ is

- (a) Constant polynomial (b) Linear polynomial
(c) Quadratic polynomial (d) Cubic polynomial

Ans: (d) Cubic polynomial

4. If $x^{51} + 51$ is divided by $x + 1$, then the remainder is

- (a) 0 (b) 1 (c) 49 (d) 50

Ans: (d) 50

$$p(x) = x^{51} + 51 \Rightarrow p(-1) = (-1)^{51} + 51 = -1 + 51 = 50$$

5. The zero of the polynomial $2x + 5$ is

- (a) $5/2$ (b) $-5/2$ (c) $2/5$ (d) $-2/5$

Ans: (b) $-5/2$

$$2x + 5 = 0 \Rightarrow 2x = -5 \Rightarrow x = -5/2$$

6. The sum of the polynomials $p(x) = x^3 - x^2 - 2$, $q(x) = x^2 - 3x + 1$

- (a) $x^3 - 3x - 1$ (b) $x^3 + 2x^2 - 1$
(c) $x^3 - 2x^2 - 3x$ (d) $x^3 - 2x^2 + 3x - 1$

Ans: (a) $x^3 - 3x - 1$

$$p(x) + q(x) = (x^3 - x^2 - 2) + (x^2 - 3x + 1) \\ = x^3 - x^2 - 2 + x^2 - 3x + 1 = x^3 - 3x - 1$$

7. Degree of the polynomial $(y^3 - 2)(y^3 + 1)$ is

- (a) 9 (b) 2 (c) 3 (d) 6

Ans: (d) 6

$$(y^3 - 2)(y^3 + 1) = y^6 + y^3 - 2y^3 - 2 = y^6 - y^3 - 2$$

8. Let the polynomials be

(A) $-13q^5 + 4q^2 + 12q$

(B) $(x^2 + 4)(x^2 + 9)$

(C) $4q^8 - q^6 + q^2$

(D) $-(5/7)y^{12} + y^3 + y^5$

Then ascending order of their degree is

(a) A, B, D, C

(b) A, B, C, D

(c) B, C, D, A

(d) B, A, C, D

Ans: (d) B, A, C, D

9. If $p(a) = 0$ then $(x - a)$ is a of $p(x)$

(a) Divisor

(b) Quotient

(c) Remainder

(d) Factor

Ans: (d) Factor

10. Zeroes of $(2 - 3x)$ is

(a) 3

(b) 2

(c) $2/3$

(d) $3/2$

Ans: (c) $2/3$

11. Which of the following has $x - 1$ as a factor?

(a) $2x - 1$

(b) $3x - 3$

(c) $4x - 3$

(d) $3x - 4$

Ans: (b) $3x - 3$

12. If $x - 3$ is a factor of $p(x)$, then the remainder is

(a) 3

(b) -3

(c) $p(3)$

(d) $p(-3)$

Ans: (c) $p(3)$

13. $(x + y)(x^2 - xy + y^2)$ is equal to

(a) $(x + y)^3$

(b) $(x - y)^3$

(c) $x^3 + y^3$

(d) $x^3 - y^3$

Ans: (c) $x^3 + y^3$

14. $(a + b - c)^2$ is equal to

(a) $(a - b + c)^2$

(b) $(-a - b + c)^2$

(c) $(a + b + c)^2$

(d) $(a - b - c)^2$

Ans: (b) $(-a - b + c)^2$

$$(-a - b + c)^2 = (-1)^2 (a + b - c)^2 = (a + b - c)^2$$

15. In an expression $ax^2 + bx + c$ the sum and product of the factors respectively ____.

(a) a, b, c

(b) b, a, c

(c) a, c, b

(d) b, c, a

Ans: (b) b, ac

16. If $(x + 5)$ and $(x - 3)$ are the factors of $ax^2 + bx + c$, then values of a, b and c are

_____.

(a) 1, 2, 3

(b) 1, 2, 15

(c) 1, 2, -15

(d) 1, -2, 15

Ans: (c) 1, 2, -15

$$(x + 5)(x - 3) = x^2 + (5 - 3)x + (5)(-3) = x^2 + 2x - 15$$

compare with $ax^2 + bx + c$, We get $a = 1, b = 2$ and $c = -15$

17. Cubic polynomial may have maximum of linear factors.

(a) 1

(b) 2

(c) 3

(d) 4

Ans: (c) 3

18. Degree of the constant polynomial is

- (a) 3 (b) 2 (c) 1 (d) 0

Ans: (d) 0

19. Find the value of m from the equation $2x + 3y = m$. If it's one solution is $x = 2$ and $y = -2$.

- (a) 2 (b) -2 (c) 10 (d) 0

Ans: (b) -2

$$2x + 3y = m \Rightarrow 2(2) + 3(-2) = m \Rightarrow 4 - 6 = m \Rightarrow -2 = m$$

20. Which of the following is a linear equation?

- (a) $x + (1/2) = 2$ (b) $x(x - 1) = 2$
(c) $3x + 5 = 2/3$ (d) $x^3 - x = 5$

Ans: (c) $3x + 5 = 2/3$

21. Which of the following is a solution of the equation $2x - y = 6$?

- (a) (2, 4) (b) (4, 2) (c) (3, -1) (d) (0, 6)

Ans: (b) (4, 2)

$$2x - y = 6 \Rightarrow 2(4) - 2 = 6 \Rightarrow 8 - 2 = 6 \Rightarrow \therefore (4, 2)$$

22. If (2, 3) is a solution of linear equation $2x + 3y = k$ then, the value of k is

- (a) 12 (b) 6 (c) 0 (d) 13

Ans: (d) 13

$$2x + 3y = k \Rightarrow 2(2) + 3(3) = k \Rightarrow 4 + 9 = k \Rightarrow 13 = k$$

23. Which condition does not satisfy the linear equation $ax + by + c = 0$

- (a) $a \neq 0, b = 0$ (b) $a = 0, b \neq 0$
(c) $a = 0, b = 0, c \neq 0$ (d) $a \neq 0, b \neq 0$

Ans: (c) $a = 0, b = 0, c \neq 0$

24. Which of the following is not a linear equation in two variable?

- (a) $ax + by + c = 0$ (b) $0x + 0y + c = 0$
(c) $0x + by + c = 0$ (d) $ax + 0y + c = 0$

Ans: (b) $0x + 0y + c = 0$

25. The value of k for which the pair of linear equations $4x + 6y - 1 = 0$ and $2x + ky - 1 = 0$ represents parallel lines is

- (a) $k = 3$ (b) $k = 2$ (c) $k = 4$ (d) $k = -3$

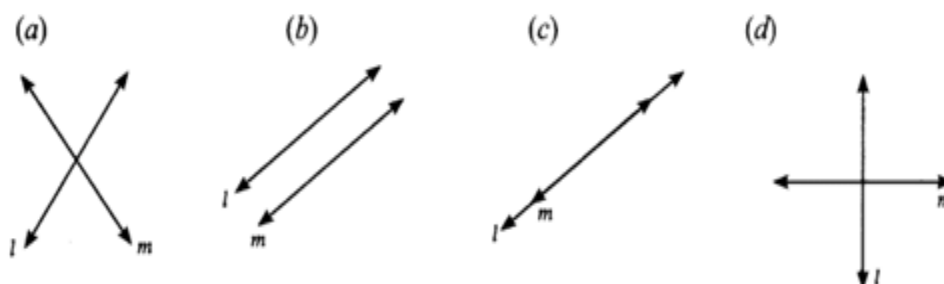
Ans: (a) $k = 3$

26. If $a_1 / a_2 \neq b_1 / b_2$ where $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ then the given pair of linear equation has Solutions.

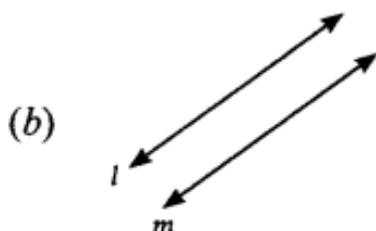
- (a) No (b) Two (c) Unique (d) Infinite

Ans: (c) Unique

27. A pair of linear equations has no solution then the graphical representation is ____.



Ans:



28. $a_1/a_2 = b_1/b_2 \neq c_1/c_2$ where $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ then the given pair of linear equation has Solutions.

- (a) No (b) Two (c) Infinite (d) Unique

Ans: (a) No

29. GCD of any two prime numbers is

- (a) -1 (b) 0 (c) 1 (d) 2

Ans: (c) 1

30. The GCD of $x^4 - y^4$ and $x^2 - y^2$ is

- (a) $x^4 - y^4$ (b) $x^2 - y^2$ (c) $(x + y)^2$ (d) $(x + y)^4$

Ans: (b) $x^2 - y^2$

$$\begin{array}{l} x^4 - y^4 = (x^2)^2 - (y^2)^2 \Rightarrow (x^2 + y^2)(x^2 - y^2) \\ x^2 - y^2 = (x^2 - y^2) \Rightarrow \text{G.C.D.} = x^2 - y^2 \end{array}$$

31. $x^2 - 2x + 1$ is a polynomial in:

- (a) One Variable (b) Two Variables (c) Three variable (d) None of the above

Ans: (a) One Variable

32. The coefficient of x^2 in $3x^3 + 2x^2 - x + 1$ is:

- (a) 1 (b) 2 (c) 3 (d) -1

Ans: (b) 2

33. A binomial of degree 20 in the following is:

- (a) $20x + 1$ (b) $x/20 + 1$ (c) $x^{20} + 1$ (d) $x^2 + 20$

Ans: (c) $x^{20} + 1$

34. The degree of $4x^3 - 12x^2 + 3x + 9$ is

- (a) 0 (b) 1 (c) 2 (d) 3

Ans: (d) 3

35. $x^2 - x$ is _____ polynomial.
 (a) Linear (b) Quadratic (c) Cubic (d) None of the above
Ans: (b) Quadratic

36. $x - x^3$ is a _____ polynomial.
 (a) Linear (b) Quadratic (c) Cubic (d) None of the above
Ans: Cubic

37. $1 + 3x$ is a _____ polynomial.
 (a) Linear (b) Quadratic (c) Cubic (d) None of the above
Ans: (a) Linear

38. The value of $f(x) = 5x - 4x^2 + 3$ when $x = -1$, is:
 (a) 3 (b) -12 (c) -6 (d) 6
Ans: (c) -6

$$f(x) = 5x - 4x^2 + 3 \Rightarrow f(-1) = 5(-1) - 4(-1)^2 + 3 = -5 - 4 + 3 = -6$$

39. The value of $p(t) = 2 + t + 2t^2 - t^3$ when $t = 0$ is
 (a) 2 (b) 1 (c) 4 (d) 0
Ans: (a) 2

40. The zero of the polynomial $f(x) = 2x + 7$ is
 (a) $2/7$ (b) $-2/7$ (c) $7/2$ (d) $-7/2$
Ans: (d) $-7/2$

$$f(x) = 2x + 7 \Rightarrow 2x + 7 = 0 \Rightarrow 2x = -7 \Rightarrow x = -7/2$$

41. What is the degree of the polynomial $\sqrt{3}$?
 (a) 0 (b) 1 (c) $1/2$ (d) 2
Ans: (a) 0

42. The degree of the constant polynomial is
 (a) 0 (b) 1 (c) 2 (d) 3
Ans: (a) 0

43. One of the linear factors of $3x^2 + 8x + 5$ is
 (a) $(x + 1)$ (b) $(x - 2)$ (c) $(x + 2)$ (d) $(x - 4)$
Ans: (a) $(x + 1)$

$$3x^2 + 8x + 5 = 3x^2 + 3x + 5x + 5 = 3x(x + 1) + 5(x + 1) = (3x + 5)(x + 1)$$

44. The coefficient of x in $7x^2 + 6x - 2$ is
 (a) 2 (b) 6 (c) -2 (d) 7
Ans: (b) 6

45. Which of the following is an example of the quadratic polynomial?
 (a) $7x + 3$ (b) $2x^2 + x - 1$ (c) $x + 3x^3 - 9$ (d) None of the above
Ans: (b) $2x^2 + x - 1$

46. Find the value of $7^2 - 5^2$

- (a) 22 (b) 23 (c) 24 (d) 25

Ans: (c) 24

$$7^2 - 5^2 = 49 - 25 = 24$$

47. If $x^2 + kx + 6 = (x + 2)(x + 3)$ for all k , find the value of k .

- (a) -1 (b) 1 (c) 3 (d) 5

Ans: (d) 5

$$x^2 + kx + 6 = (x + 2)(x + 3) = x^2 + 3x + 2x + 6 = x^2 + 5x + 6 \Rightarrow k = 5$$

48. What is the zero of the polynomial $p(x) = cx + d$?

- (a) $-c$ (b) $-d$ (c) $-d / c$ (d) d / c

Ans: (c) $-d / c$

49. The zero of the polynomial $p(x) = -5x + 5$ is

- (a) 0 (b) -5 (c) -1 (d) 1

Ans: (d) 1

50. Which of the following is a constant polynomial?

- (a) $4x + 1$ (b) 3 (c) $2x^2$ (d) $6x + 3$

Ans: (b) 3

51. The linear equation $3x - 11y = 10$ has:

- (a) Unique solution (b) Two solutions
(c) Infinitely many solutions (d) No solutions

Ans: (c) Infinitely many solutions

52. $3x + 10 = 0$ will have _____ solution / solutions.

- (a) Unique (b) Two (c) Infinitely many (d) No

Ans: (a) Unique

$$3x + 10 = 0 \Rightarrow x = -10 / 3 \Rightarrow \text{Hence, only one solution is possible.}$$

53. The solution of equation $x - 2y = 4$ is:

- (a) (0, 2) (b) (2, 0) (c) (4, 0) (d) (1, 1)

Ans: (c) (2, 0)

54. Find the value of k , if $x = 1$, $y = 2$ is a solution of the equation $2x + 3y = k$.

- (a) 5 (b) 6 (c) 7 (d) 8

Ans: (d) 8

$$2x + 3y = k \Rightarrow k = 2(1) + 3(2) = 2 + 6 = 8$$

55. Point (3, 4) lies on the graph of the equation $3y = kx + 7$. The value of k is:

- (a) $4 / 3$ (b) $5 / 3$ (c) 3 (d) $7 / 3$

Ans: (b) $5 / 3$

$$3y = kx + 7 \Rightarrow (3 \times 4) = (k \times 3) + 7 \Rightarrow 12 = 3k + 7 \Rightarrow 3k = 5 \Rightarrow k = 5 / 3$$

56. The graph of linear equation $x + 2y = 2$, cuts the y-axis at:
(a) (2, 0) (b) (0, 2) (c) (0, 1) (d) (1, 1)

Ans: (c) (0, 1)

$$x + 2y = 2 \Rightarrow \text{If } x=0, \text{ then; } y = 1$$

57. Any point on line $x = y$ is of the form:
(a) (k, -k) (b) (0, k) (c) (k, 0) (d) (k, k)

Ans: (d) (k, k)

58. The graph of $x = 3$ is a line:

- (a) Parallel to the x-axis at a distance of 3 units from the origin
(b) Parallel to the y-axis at a distance of 3 units from the origin
(c) Makes an intercept 3 on the x-axis
(d) Makes an intercept 3 on the y-axis

Ans: (b) Parallel to the y-axis at a distance of 3 units from the origin

59. In equation, $y = mx + c$, m is:

- (a) Intercept (b) Slope (c) Solution of the equation (d) None of the above

Ans: (b) Slope

60. If x and y are both positive solutions of equation $ax + by + c = 0$, always lie in the:

- (a) First quadrant (b) Second quadrant
(c) Third quadrant (d) Fourth quadrant

Ans: (a) First quadrant

61. A linear equation in two variables is of the form $ax + by + c = 0$, where

- (a) $a = 0, c = 0$ (b) $a \neq 0, b = 0$ (c) $a = 0, b \neq 0$ (d) $a \neq 0, b \neq 0$

Ans: (d) $a \neq 0, b \neq 0$

62. Any point on the x-axis is of the form

- (a) (x, y) (b) (0, y) (c) (x, 0) (d) (x, x)

Ans: (c) (x, 0)

63. Any point on the y-axis is of the form

- (a) (y, y) (b) (0, y) (c) (x, y) (d) (x, 0)

Ans: (b) (0, y)

64. The linear equation $2x - 5y = 7$ has _____ solution / solutions.

- (a) No (b) Unique (c) Two (d) Infinitely many

Ans: (d) Infinitely many

65. The linear equation $3x - y = x - 1$ has _____ solution / solutions.

- (a) No (b) Unique (c) Two (d) Infinitely many

Ans: d. Infinitely many

66. The graph of the linear equation $2x + 3y = 6$ cuts the y-axis at the point

- (a) (2, 0) (b) (0, 2) (c) (3, 0) (d) (0, 3)

Ans: (b) (0, 2)

67. The equation $2x + 5y = 7$ has a unique solution, if x, y are:

- (a) Rational numbers (b) Real numbers
(c) Natural numbers (d) Positive real numbers

Ans: (c) Natural numbers

68. The point of the form (a, a) always lies on:

- (a) On the line $x + y = 0$ (b) On the line $y = x$
(c) x - axis (d) y - axis

Ans: (b) On the line $y = x$

69. If we multiply or divide both sides of a linear equation with the same non-zero number, then the solution of the linear equation:

- (a) Remains the same (b) Changes in addition only
(c) Changes in case of multiplication only (d) Changes in case of division only

Ans: (a) Remains the same

70. If $(2, 0)$ is a solution of the linear equation $2x + 3y = k$, then the value of k is

- (a) 2 (b) 4 (c) 5 (d) 6

Ans: (b) 4

71. $x - 1$ is a factor of $4x^2 - 9x - 6$.

- (a) True (b) False ((c) can't determine ((d) None of these

Ans: (b) False

72. Find the value of k , if $x - 3$ is a factor of $5x^3 - 2x^2 + x + k$.

- (a) 50 (b) 60 (c) -60 (d) -120

Ans: (d) - 120

$$p(3) = 5(3)^3 - 2(3)^2 + 3 + k = 0 \Rightarrow 135 - 18 + 3 + k = 0 \Rightarrow 120 + k = 0 \Rightarrow k = -120$$

73. What do we get after factorising $x^2 + 6x - 27$?

- (a) $(x + 9)(x - 3)$ (b) $(x - 9)(x + 3)$ (c) $(x - 9)(x - 3)$ (d) $(x + 9)(x + 3)$

Ans: (a) $(x + 9)(x - 3)$

74. What do we get after factoring $49x^2 - 28xy + 4y^2$?

- (a) $(7x + 2y)^2$ (b) $(49x - 4y)^2$ (c) $(7x - 2y)^2$ (d) $(7x - 28y)^2$

Ans: (c) $(7x - 2y)^2$

$$a^2 - 2ab + b^2 = (a - b)^2 \quad (\text{Algebraic Identity})$$
$$\sqrt{49x^2} - 28xy + \sqrt{4y^2} = (7x - 2y)^2$$

75. $26 * 34 =$ _____ (calculate without direct calculation).

- (a) 900 (b) 884 (c) 916 (d) 844

Ans: (b) 884

$$(a - b) * (a + b) = a^2 - b^2 \quad (\text{Algebraic identity})$$
$$26 * 34 = (30 - 4) * (30 + 4) = 30^2 - 4^2 = 900 - 16 = 884.$$

76. $27 \times 29 =$ _____ (calculate without direct calculation).

- (a) 783 (b) 753 (c) 763 (d) 793

Ans: (a) 783

$$(a - b)(a + b) = a^2 - b^2 \text{ (Algebraic identity)}$$

$$(28 - 1) \times (28 + 1) = 28^2 - 1^2 = 784 - 1 = 783$$

77. $95^3 =$ _____ (calculate without direct calculation).

- (a) 856395 (b) 857625 (c) 857375 (d) 852395

Ans: (c) 857375

$$(x - y)^3 = x^3 - y^3 - 3xy(x - y) \text{ (Algebraic Identity)}$$

$$95^3 = (100 - 5)^3 = (100)^3 - (5)^3 - 3(100)(5)(100 - 5)$$

$$= 1000000 - 125 - 1500(95) = 1000000 - 125 - 142500 = 857375$$

78. What do we get after expanding $(p + 3q - 2z)^2$?

- (a) $p^2 + 9q^2 + 4z^2 + 6pq - 12qz + 4zp$
 (b) $p^2 + 9q^2 + 4z^2 + 12pq + 12qz + 4zp$
 (c) $p^2 + 9q^2 + 4z^2 - 12pq - 12qz - 4zp$
 (d) $p^2 + 9q^2 + 4z^2 + 6pq - 12qz - 4zp$

Ans: (d) $p^2 + 9q^2 + 4z^2 + 6pq - 12qz - 4zp$

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca \text{ (Algebraic identity)}$$

$$= p^2 + (3q)^2 + (-2z)^2 + 2(p)(3q) + 2(3q)(-2z) + 2(-2z)(p)$$

$$= p^2 + 9q^2 + 4z^2 + 6pq - 12qz - 4zp.$$

79. What do we get after factorising $x^3 + 8y^3 + z^3 - 4xyz$?

- (a) $(x + 2y - z)(x^2 + 4y^2 + z^2 - 2xy - 2yz - zx)$
 (b) $(x + 2y + z)(x^2 + 4y^2 + z^2 - 2xy + 2yz - zx)$
 (c) $(x + 2y + z)(x^2 + 4y^2 + z^2 - 2xy - 2yz - zx)$
 (d) $(x + 2y + z)(x^2 + 4y^2 + z^2 - 2xy - 2yz + zx)$

Ans: (c) $(x + 2y + z)(x^2 + 4y^2 + z^2 - 2xy - 2yz - zx)$

$$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$x^3 + 8y^3 + z^3 - 4xyz = (x + 2y + z)(x^2 + (2y)^2 + z^2 - x(2y) - (2y)(z) - zx)$$

$$= (x + 2y + z)(x^2 + 4y^2 + z^2 - 2xy - 2yz - zx).$$

80. What is the value of $(a + b)^2$?

- (a) $a^2 + b^2$ (b) $a^2 + 2ab + b^2$ (c) $a^2 - 2ab + b^2$ (d) $a^2 + ab + b^2$

Ans: (b) $a^2 + 2ab + b^2$

81. Which of the following is the factorization of $x^2 - y^2$?

- (a) $(x - y)^2$ (b) $(x + y)^2$ (c) $(x - y)(x + y)$ (d) $x^2 + y^2$

Ans: (c) $(x - y)(x + y)$

82. The expansion of $(2x - 3)^2$ is:

- (a) $4x^2 - 9$ (b) $4x^2 - 12x + 9$ (c) $4x^2 + 12x + 9$ (d) $4x^2 - 6x + 9$

Ans: (b) $4x^2 - 12x + 9$

83. What is the value of $(a - b)^3$ expanded?

- (a) $a^3 - b^3$ (b) $a^3 - 3a^2b + 3ab^2 - b^3$
(c) $a^3 + 3a^2b + 3ab^2 + b^3$ (d) $a^3 - 3ab + b^3$

Ans: (b) $a^3 - 3a^2b + 3ab^2 - b^3$

84. The identity $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ is an example of:

- (a) Factorization of sum of cubes
(b) Factorization of difference of cubes
(c) Expansion of binomial square
(d) None of the above

Ans: (a) Factorization of sum of cubes

85. Simplify $(x + 3)(x - 3)$:

- (a) $x^2 + 9$ (b) $x^2 - 9$ (c) $x^2 - 6x + 9$ (d) $x^2 + 6x + 9$

Ans: (b) $x^2 - 9$

86. The expansion of $(x + y)^3$ is:

- (a) $x^3 + y^3$ (b) $x^3 + 3x^2y + 3xy^2 + y^3$
(c) $x^3 - 3x^2y + 3xy^2 - y^3$ (d) $x^3 + 3xy + y^3$

Ans: (b) $x^3 + 3x^2y + 3xy^2 + y^3$

87. Which of the following is the expansion of $(a + b)(a - b)$?

- (a) $a^2 + b^2$ (b) $a^2 - b^2$ (c) $a^2 - 2ab + b^2$ (d) $a^2 + 2ab + b^2$

Ans: (b) $a^2 - b^2$

88. Factorize $x^2 + 5x + 6$:

- (a) $(x + 2)(x + 3)$ (b) $(x - 2)(x - 3)$ (c) $(x + 1)(x + 6)$ (d) $(x - 1)(x - 6)$

Ans: (a) $(x + 2)(x + 3)$

89. The identity $(x + y)^2 = x^2 + 2xy + y^2$ is called:

- (a) Cubic identity (b) Square of a binomial
(c) Difference of squares (d) Sum of cubes

Ans: (b) Square of a binomial

90. A polynomial with one degree is called _____ polynomial.

- (a) Linear (b) Quadratic (c) Monomial (d) Binomial

Ans: (a) Linear polynomial

91. If $x + 2$ is a factor of $x^3 - 2ax^2 + 16$, then value of a is

- (a) 3 (b) 1 (c) 4 (d) 2

Ans: (b) 1

92. If one of the factor of $x^2 + x - 20$ is $(x + 5)$. Find the other

- (a) $x - 4$ (b) $x + 2$ (c) $x + 4$ (d) $x - 5$

Ans: (a) $x - 4$

93. The value of $99^2 - 98^2$ is: _____ (calculate without direct calculation)

- (a) 1 (b) 197 (c) 187 (d) 207

Ans: (b) 197

$$a^2 - b^2 = (a - b)(a + b) \quad (\text{Algebraic Identity})$$

$$99^2 - 98^2 = (99 - 98)(99 + 98) = 1 \times 197 = 197$$

94. One of the factors of $(1 + 7x)^2 + (49x^2 - 1)$ is:

- (a) $x - 7$ (b) $7 - x$ (c) $7x - 1$ (d) $14x$

Ans: (d) $14x$

$$(1 + 7x)[(1 + 7x) + (7x - 1)] \Rightarrow (1 + 7x)[1 + 7x + 7x - 1] \Rightarrow (1 + 7x)[14x]$$

95. The factorization of $6x^2 + 11x + 3$ is:

- (a) $(3x + 1)(2x + 3)$ (b) $(x + 1)(2x + 3)$
(c) $(x + 3)(2x + 1)$ (d) $(3x + 3)(x + 1)$

Ans: (a) $(3x + 1)(2x + 3)$

96. If $y^{97} + 97$ is divided by $y + 1$, the remainder is:

- (a) 0 (b) 1 (c) 95 (d) 96

Ans: (d) 96

$$f(y) = y^{97} + 97 \Rightarrow y = -1: f(-1) = (-1)^{97} + 97 = 96$$

97. $\sqrt{12} \times \sqrt{15}$ is equal to:

- (a) $5\sqrt{6}$ (b) $6\sqrt{5}$ (c) $10\sqrt{5}$ (d) $\sqrt{25}$

Ans: (b) $6\sqrt{5}$

$$\sqrt{12} \times \sqrt{15} = \sqrt{12 \times 15} = \sqrt{180} = \sqrt{36 \times 5} = 6 \times \sqrt{5} = 6\sqrt{5}$$

98. The zero of the polynomial $p(x) = -9x + 9$ is:

- (a) 0 (b) -9 (c) -1 (d) 1

Ans: (d) 1

99. The value of the polynomial $7x^4 + 3x^2 - 4$, when $x = -2$ is:

- (a) 100 (b) 110 (c) 120 (d) 130

Ans: (c) 120

100. Degree of the polynomial $7x^5 + 8x^2 - 5x + 3$ is:

- (a) 1 (b) 3 (c) 2 (d) 5

Ans: (d) 5